

```
$ psql -c "SELECT ... FROM students JOIN courses ON ..."
```

# SQL JOIN

// 0000000000000000

INNER

□□

LEFT

□□□□

RIGHT

□□□□

FULL

□□

CROSS

□□□□

\$ cat query-plan.md



**01 /**   **JOIN**

□□□□□□□□□□□□□□

**02 /**  

□□□□□□□

**03 /**    

ON vs WHERE □□□

**04 /**  

□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□ = LEFT□□□ = INNER□

## PART 01

NO GLYPH | NO GLYPH | JOIN

🐼 ID: 1,2,3,4 × 🐼 ID: 2,3,5

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

```
1 SELECT
2     s.student_id,
3     s.name AS student_name,
4     c.course_name
5 FROM students AS s
6 INNER JOIN courses AS c
7     ON s.student_id = c.student_id;
```

[illegible]

## RESULT SET

### ID1, 2, 3, 4

### ID 2, 3, 5

→  $\mathbb{R}^2$  ID 2, 3

$x \approx 1,4000$

**x** 00000005000000

# LEFT JOIN

NO GRAPH



```
1  SELECT
2      s.name,
3      c.course_name,
4      CASE
5          WHEN c.course_name IS NULL
6          THEN 'NULL' ELSE 'NULL'
7      END AS status
8  FROM students AS s
9  LEFT JOIN courses AS c
10     ON s.student_id = c.student_id;
```

## RESULT SET

→ ID 1, 2, 3, 4

→ 1 4 NULL



# LEFT JOIN □□□□□

NOT RUN NOT RUN 1 NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN

```
1 SELECT c.customer_name
2 FROM customers c
3 LEFT JOIN orders o
4     ON c.customer_id = o.customer_id
5 WHERE o.order_id IS NULL;
6 -- □□□□□□
```

NOT RUN NOT RUN 2 NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN NOT RUN

```
1 SELECT s.name,
2         COUNT(c.course_id) AS course_count
3 FROM students s
4 LEFT JOIN courses c
5     ON s.student_id = c.student_id
6 GROUP BY s.student_id, s.name;
7 -- □□□□□□ count = 0
```

LEFT JOIN + IS NULL □□□□□□□□□□—INNER JOIN □□□

# RIGHT FULL OUTER CROSS

## RIGHT JOIN

□□□□□□□□□□

A LEFT JOIN B ≡ B RIGHT JOIN

A □ □ □ LEFT □ □ □

## FULL OUTER JOIN

□ □ □ □ □ □ □ □ □ □ NULL □

MySQL □ □ □ □ □ □ □ □ LEFT UNION RIGHT

□ UNION □ □ □

## CROSS JOIN

□ □ □ □ □ □ □ □ 3 × 4 □ □ = 12 □ □ □

□ □ □ □ □ □ □ □ □ □ × □ □ □ □ □ □ □ □ □ □ □ □

```
1  -- MySQL 实现 FULL OUTER JOIN
2  SELECT * FROM students s LEFT JOIN courses c ON s.student_id = c.student_id
3  UNION
4  SELECT * FROM students s RIGHT JOIN courses c ON s.student_id = c.student_id;
```

## PART 02



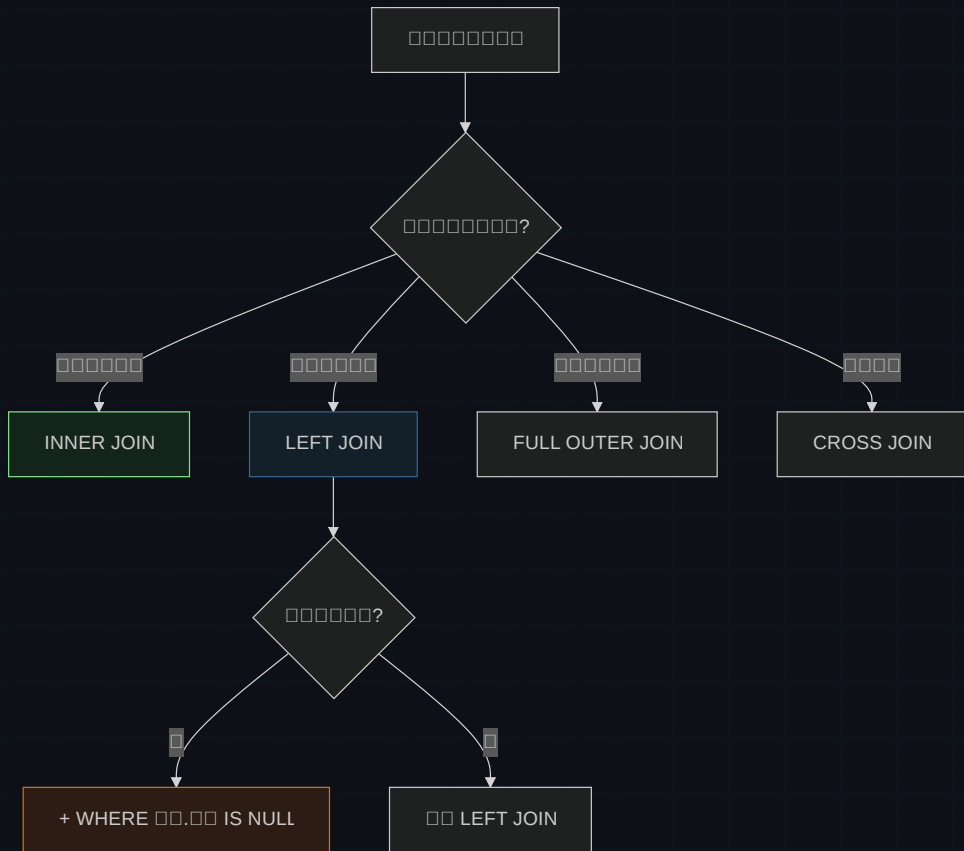
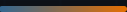
$$\square\square\square + \square\square\square\square$$



# JOIN

| JOIN       | Table 1 | Table 2 | Result     | Performance |
|------------|---------|---------|------------|-------------|
| INNER JOIN | 10000   | 10000   | 1000       | ★★★★★       |
| LEFT JOIN  | 100     | 10000   | 10000 NULL | ★★★★★       |
| RIGHT JOIN | 10000   | 100     | 100 NULL   | ★★          |
| FULL OUTER | 100     | 100     | 10000 NULL | ★★          |
| CROSS JOIN | 100     | 100     | 1000000    | ★           |

\$ 90% 1000000 INNER 100 LEFT — 100000000



PART 03

# □□□□□ ON vs WHERE

90% ⚡ LEFT JOIN bug □□□

CHECKPOINT\_01

LEFT JOIN courses c ... WHERE c.course\_name = 'Math' □□□□

A □□□□□□□□□□ Math □□□ NULL

B □□□□ Math □□□—LEFT JOIN □ WHERE □□ INNER □□

C □□□□□LEFT JOIN □□□□□ WHERE



```
1  -- ❌ WHERE JOIN 查询 → NULL 查询 INNER 查询
2  SELECT * FROM students s
3  LEFT JOIN courses c ON s.student_id = c.student_id
4  WHERE c.course_name = 'Math'; -- 查询 Math 课程
5
6  -- ✅ ON JOIN 查询 → 查询
7  SELECT * FROM students s
8  LEFT JOIN courses c
9      ON s.student_id = c.student_id
10     AND c.course_name = 'Math'; -- 查询 Math 课程 NULL
11
12 -- ✅ 查询 → IS NULL 查询 WHERE
13 SELECT * FROM students s
14 LEFT JOIN courses c ON s.student_id = c.student_id
15 WHERE c.student_id IS NULL;
```

ON = JOIN 查询 WHERE = JOIN 查询

## PART 04





```
1  -- 0000000000000000
2  -- students ← enrollments → courses
3
4  SELECT
5      s.name AS student_name,
6      c.course_name,
7      e.enrollment_date
8  FROM students s
9  INNER JOIN enrollments e ON s.student_id = e.student_id
10 INNER JOIN courses c ON e.course_id = c.course_id;
```

00000

UNIQUE KEY (student\_id, course\_id) 00000

 JOIN

 INNER JOIN 0000000000000000

NO GRAPH  
NO GRAPH  
NO GRAPH  
NO GRAPH  
NO GRAPH

# JOIN □□□

NO GRAPH  
NO GRAPH  
1  
NO GRAPH  
NO GRAPH  
NO GRAPH  
ON

```
1  -- ✖ 遅い
2  SELECT * FROM students s
3  INNER JOIN courses c;
4  -- 遅い = 遅い × 遅い
```

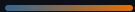
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2  
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```
1  -- ✔ 速い
2  CREATE INDEX idx_courses_student_id
3  ON courses(student_id);
```

## DEBUG FLOW

1. EXPLAIN 実行 → 2. ON 実行 → 3. 実行時間 → 4. JOIN 実行 5. 実行時間



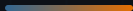


DO

- ✓ `students AS s`
- ✓ `SELECT *`
- ✓ `LEFT JOIN` `RIGHT JOIN`

DON'T

- × `ON` →
- × `LEFT` `INNER` →
- × `WHERE` → `LEFT` `INNER`



INNER = □□

□□□□□□□□

LEFT = □□

NOT NULL NOT NULL NOT NULL NOT NULL + IS NULL □□□□□

ON vs WHERE

JOIN NOT NULL NOT NULL NOT NULL NOT NULL vs JOIN □□□

□□

NOT NULL NOT NULL NOT NULL NOT NULL NOT NULL NOT NULL NOT NULL NOT NULL ON □□

□□□□□□□□□□□□□□□□ — □ = LEFT□□□ = INNER□

# Tables joined.

⚡⚡⚡⚡⚡⚡⚡⚡ JOIN ⚡⚡

```
$ EXPLAIN SELECT ... FROM a LEFT JOIN b ON ...  
inner → left → on/where → index
```